1 $3(3x-1) = 4(x+5)$ or $9x - 3 = 4x + 20$ **[M1]**

$5x = 23$ **[M1]**

4.6 or $\frac{23}{5}$ **[A1]**

[Total Marks: 3]

2 Valid method to eliminate one variable, e.g. multiplying the first equation by 2 to get $4x - 2y = 14$ and adding the equations **[M1]**

$x = 4$ **[A1]**

$y = 1$ **[A1]**

[Total Marks: 3]

3 Multiplies one or both equations to equate coefficients of x or y (e.g. $4x - 10y = 44$) **[M1]**

Adds or subtracts equations to eliminate one variable (e.g. $13y = -39$) **[M1]**

$x = 3.5$, $y = -3$ **[A1]**

[Total Marks: 3]

4 (a) $4x - 12 = 2x + 10$ **[M1]**

$x = 11$ **[A1]**

(b) $5y = 2x - 1$ **[M1]**

$\frac{5y+1}{2}$ **[A1]**

[Total Marks: 4]

5 (a) $6x - 3 = 5x + 8$ **[M1]**

11 **[A1]**

(b) $y + 7 = 4x$ or $\frac{y}{4} = x - \frac{7}{4}$ **[M1]**

$\frac{y+7}{4}$ or $\frac{y}{4} + \frac{7}{4}$ **[A1]**

[Total Marks: 4]

6 (a) $6x - 3 = 5x + 8$ **[M1]**

11 **[A1]**

(b) $5y = 2x + 3$ **[M1]**

$2x = 5y - 3$ **[M1]**

$\frac{5y-3}{2}$ **[A1]**

[Total Marks: 5]

7 **(a)** Expands brackets correctly to $6x - 3 = 4x + 11$ **[M1]**

7 **[A1]**

(b) Multiplies second equation by 2 to get $2x - 2y = 4$ OR substitutes $x = y + 2$ into first equation **[M1]**

Correctly finds $x = 4$ **[A1]**

Correctly finds $y = 2$ **[A1]**

[Total Marks: 5]

8 **(a)** $5x - 3 = 2x + 12$ **[M1]**

5 **[A1]**

(b) Multiply by 12 to get $4(x + 2) + 3(2x - 1) = 36$ (or equivalent common denominator method) **[M1]**

$4x + 8 + 6x - 3 = 36$ **[M1]**

$\frac{31}{10}$ or 3.1 **[A1]**

[Total Marks: 5]

9 **(a)** Expands bracket to obtain $6x - 15$ **[M1]**

Rearranges terms to isolate x, e.g., $6x - 4x = 7 + 15$ or $2x = 22$ **[M1]**

11 **[A1]**

(b) Multiplies by 5 to obtain $5y = 2x + 3$ **[M1]**

$\frac{5y - 3}{2}$ **[A1]**

[Total Marks: 5]

10 **(a)** $8x - 4 = 3x + 11$ **[M1]**

$5x = 15$ **[M1]**

3 **[A1]**

(b) $2y = 3x - 5$ **[M1]**

$2y + 5 = 3x$ **[M1]**

$\frac{2y + 5}{3}$ or $\frac{2y}{3} + \frac{5}{3}$ **[A1]**

[Total Marks: 6]

11 **(a)** $5x - 10 = 3x + 14$ **[M1]**

$2x = 24$ **[M1]**

12 **[A1]**

(b) $2A = h(a + b)$ or $\frac{A}{h} = \frac{a+b}{2}$ **[M1]**

$\frac{2A}{h} = a + b$ or $2A = ha + hb$ **[M1]**

$\frac{2A}{h} - a$ or $\frac{2A - ah}{h}$ **[A1]**

[Total Marks: 6]

12 **(a)** $6x - 15 = 4x + 7$ **[M1]**

$2x = 22$ **[M1]**

11 **[A1]**

(b) Equates coefficients, e.g. multiplies first equation by 2 and second equation by 3 to get $4x + 6y = 26$ and $15x - 6y = 12$ **[M1]**

Adds equations to eliminate y, e.g. $19x = 38$, so $x = 2$ **[A1]**

$y = 3$ **[A1]**

[Total Marks: 6]

13 **(a)** Expands brackets: $4x - 12 = 2x + 7$ **[M1]**

Rearranges to collect terms: $2x = 19$ **[M1]**

9.5 or $\frac{19}{2}$ **[A1]**

(b) Multiplies first equation by 2 to equate coefficients of y: $10x - 4y = 32$ (or equivalent valid method) **[M1]**

Adds equations to eliminate y: $13x = 39 \implies x = 3$ **[A1]**

Substitutes $x = 3$ into either equation to find y: $15 - 2y = 16 \implies y = -0.5$ **[A1]**

[Total Marks: 6]

14 $y(x + 4) = 3x - 2$ **[M1]**

$xy - 3x = -4y - 2$ or $3x - xy = 4y + 2$ **[M1]**

$x(y - 3) = -4y - 2$ or $x(3 - y) = 4y + 2$ **[M1]**

$\frac{4y+2}{3-y}$ or $\frac{-4y-2}{y-3}$ **[A1]**

[Total Marks: 4]

15 $y(x - 5) = 2x + 3$ or $xy - 5y = 2x + 3$ **[M1]**

$xy - 2x = 5y + 3$ or $2x - xy = -5y - 3$ **[M1]**

$x(y - 2) = 5y + 3$ or $x(2 - y) = -5y - 3$ **[M1]**

$\frac{5y+3}{y-2}$ or $\frac{-5y-3}{2-y}$ **[A1]**

[Total Marks: 4]

16 **(a)** Multiplies equations to equate coefficients of x or y, e.g., $9x + 6y = 36$ and $10x - 6y = 2$ **[M1]**

Adds or subtracts equations to eliminate a variable **[M1]**

$x = 2$ and $y = 3$ **[A1]**

(b) $(2x \pm a)(x \pm b) = 0$ where $ab = 12$ **[M1]**

$(2x + 3)(x - 4) = 0$ **[M1]**

$x = 4$ and $x = -1.5$ (or $-\frac{3}{2}$) **[A1]**

[Total Marks: 6]

17 **(a)** Correct method to equate coefficients (e.g., $10x + 15y = 65$ and $10x - 4y = 8$, leading to $19y = 57$) or correctly substitute **[M1]**

$y = 3$ **[A1]**

$x = 2$ **[A1]**

(b) $y(x + 2) = 3x - 1$ **[M1]**

$xy + 2y = 3x - 1$ **[M1]**

$xy - 3x = -1 - 2y$ or $2y + 1 = 3x - xy$ **[M1]**

$\frac{-1-2y}{y-3}$ or $\frac{2y+1}{3-y}$ **[A1]**

[Total Marks: 7]

18 **(a)** $y(x + 4) = 3x - 2$ **[M1]**

$xy + 4y = 3x - 2$ **[M1]**

$xy - 3x = -4y - 2$ or $x(y - 3) = -(4y + 2)$ **[M1]**

$\frac{-4y-2}{y-3}$ or $\frac{4y+2}{3-y}$ **[A1]**

(b) $5(x + 1) = 2(x - 2)$ **[M1]**

$5x + 5 = 2x - 4$ **[M1]**

-3 **[A1]**

[Total Marks: 7]

19 **(a)** Multiplies equations to equate coefficients, e.g., $9x + 6y = 36$ and $10x - 6y = 2$ **[M1]**

Adds or subtracts equations to eliminate one variable, e.g., $19x = 38$ **[M1]**

$x = 2$ **[A1]**

$y = 3$ **[A1]**

(b) Finds correct product and sum components, e.g., $-10x$ and $3x$ **[M1]**

Correctly factorises to $(2x + 3)(x - 5) = 0$ **[M1]**

$x = -1.5, x = 5$ **[A1]**

[Total Marks: 7]

20 (a) Multiply the second equation by 2 to get $8x + 2y = 36$ (or correct substitution method) **[M1]**

$x = 5$ **[A1]**

$y = -2$ **[A1]**

(b) $T(p - 2) = 3p + 4$ **[M1]**

$Tp - 2T = 3p + 4$ **[M1]**

$Tp - 3p = 2T + 4$ or $p(T - 3) = 2T + 4$ **[M1]**

$\frac{2T + 4}{T - 3}$ or $\frac{-2T - 4}{3 - T}$ **[A1]**

[Total Marks: 7]

21 (a) Multiplies the first equation by 2 to give $4x - 2y = 18$ (or rearranges to $y = 2x - 9$) **[M1]**

Adds the equations to eliminate y, yielding $7x = 28$ (or correctly substitutes to form an equation in one variable) **[M1]**

$x = 4$ **[A1]**

$y = -1$ **[A1]**

(b) Multiplies throughout by a common denominator (e.g., 12) to obtain $4(2x) - 3(x - 4) = 30$ **[M1]**

Expands brackets correctly: $8x - 3x + 12 = 30$ **[M1]**

$5x = 18$ **[M1]**

3.6 or $\frac{18}{5}$ **[A1]**

[Total Marks: 8]

22 (a) Multiply second equation by 2: $4x - 10y = 18$ **[M1]**

Subtract equations to eliminate x: $13y = -13$ **[M1]**

$y = -1$ **[A1]**

$x = 2$ **[A1]**

(b) $(3x + 2)(x - 4) = 0$ **[M2]**

If M0, M1 for $(3x \pm a)(x \pm b)$ where $ab = 8$ or $a \pm 3b = -10$

$x = -\frac{2}{3}$ **[A1]**

$x = 4$ **[A1]**

[Total Marks: 8]

23 Forms common denominator or multiplies through: $2(x-2)+1(x-1)=(x-1)(x-2)$ **[M1]**

Expands brackets correctly: $2x-4+x-1=x^2-3x+2$ **[M1]**

Rearranges to a 3-term quadratic: $x^2-6x+7=0$ **[A1]**

Applies quadratic formula: $\dfrac{-(-6)\pm\sqrt{(-6)^2-4(1)(7)}}{2(1)}$ or completes the square: $(x-3)^2-9+7=0$ **[M1]**

Correct unsimplified surd form: $\dfrac{6\pm\sqrt{8}}{2}$ or $(x-3)^2=2$ **[M1]**

$3\pm\sqrt{2}$ **[A1]**

[Total Marks: 6]

24 (a) $p(x+4)=3x-1$ **[M1]**

$px+4p=3x-1$ **[M1]**

$3x-px=4p+1$ (or equivalent step isolating x terms on one side) **[M1]**

$\dfrac{4p+1}{3-p}$ (or $\dfrac{-4p-1}{p-3}$) **[A1]**

(b) $(2x\pm a)(x\pm b)=0$ where $ab=-12$ or $2b\pm a=-5$ **[M1]**

$(2x+3)(x-4)=0$ **[M1]**

$x=-1.5$, $x=4$ **[A1]**

[Total Marks: 7]

25 (a) $2(3x+2+x-1)=42$ or $3x+2+x-1=21$ **[M1]**

$4x+1=21$ **[M1]**

5 **[A1]**

(b) $p^2=\dfrac{x-3}{x}$ **[M1]**

$p^2x=x-3$ **[M1]**

$p^2x-x=-3$ or $x-p^2x=3$ or $x(p^2-1)=-3$ **[M1]**

$\dfrac{3}{1-p^2}$ or $\dfrac{-3}{p^2-1}$ **[A1]**

[Total Marks: 7]

26 (a) Multiplies by denominator: $y(x-3)=2x+5$ **[M1]**

Expands and collects x terms on one side: $xy-2x=3y+5$ **[M1]**

$\frac{3y+5}{y-2}$ or $\frac{-3y-5}{2-y}$ **[A1]**

(b) Multiplies throughout by common denominator $3(x-2)$ to reach $(x+1)(x-2)-6=3(x-2)$ **[M1]**

Expands brackets correctly: $x^2-2x+x-2-6=3x-6$ **[M1]**

Rearranges to form a correct 3-term quadratic equation: $x^2-4x-2=0$ **[A1]**

Uses the quadratic formula $\frac{-(-4)\pm\sqrt{(-4)^2-4(1)(-2)}}{2(1)}$ **[M1]**

4.45 **[A1]**

-0.45 **[A1]**

[Total Marks: 9]

27 **(a)** Sets up equation for area: $(2x+3)(x-1)=45$ **[M1]**

Expands and simplifies to given result: $2x^2-2x+3x-3=45 \implies 2x^2+x-48=0$ **[A1]**

(b) Uses quadratic formula with correct values: $\frac{-1\pm\sqrt{1^2-4(2)(-48)}}{2(2)}$ **[M1]**

Correct discriminant: $\frac{-1\pm\sqrt{385}}{4}$ **[M1]**

4.655... and $-5.155...$ **[A1]**

4.66 and -5.16 **[A1]**

(c) Multiplies by denominator: $y(x-3)=ax+5$ **[M1]**

Expands brackets: $xy-3y=ax+5$ **[M1]**

Collects terms in x on one side: $xy-ax=3y+5$ **[M1]**

Factorises x and divides: $\frac{3y+5}{y-a}$ (or equivalent) **[A1]**

[Total Marks: 10]

28 $\frac{3(x-1)+1(x+2)}{(x+2)(x-1)}=2$ or $3(x-1)+1(x+2)=2(x+2)(x-1)$ **[M1]**

$3x-3+x+2$ or $4x-1$ **[M1]**

$2x^2-2x-3=0$ or $-2x^2+2x+3=0$ **[A2]**

$\frac{-(-2)\pm\sqrt{(-2)^2-4(2)(-3)}}{2(2)}$ or GDC equivalent to solve the 3-term quadratic **[M1]**

1.82 and -0.82 **[A1]**

[Total Marks: 6]

29 Multiplies by the common denominator to get $3(2x+1)+4(x-2)=2(x-2)(2x+1)$ [M2]
(allow M1 for $\frac{3(2x+1)+4(x-2)}{(x-2)(2x+1)}=2$)

Expands brackets to get $10x - 5 = 2(2x^2 - 3x - 2)$ or $10x - 5 = 4x^2 - 6x - 4$ **[M1]**

Rearranges to form a correct 3-term quadratic equation equal to zero: $4x^2 - 16x + 1 = 0$ **[A1]**

Substitutes correctly into the quadratic formula: $x = \dfrac{-(-16) \pm \sqrt{(-16)^2 - 4(4)(1)}}{2(4)}$ **[M1]**

$x = 3.94$ and $x = 0.06$ **[A1]**

[Total Marks: 6]

30 Multiplies by the common denominator to get $5(2x+1) - 3(x-2) = 2(x-2)(2x+1)$ **[M2]**

If not M2, M1 for a correct single fraction $\dfrac{5(2x+1) - 3(x-2)}{(x-2)(2x+1)}$ seen

$4x^2 - 13x - 15 = 0$ **[A2]**

If not A2, A1 for correctly expanding sides to $7x + 11$ or $4x^2 - 6x - 4$

$x = \dfrac{-(-13) \pm \sqrt{(-13)^2 - 4(4)(-15)}}{2(4)}$ **[M1]**

4.15 **[A1]**

-0.90 **[A1]**

[Total Marks: 7]

31 (a) $\dfrac{3(x+3) - 2(x-1)}{(x-1)(x+3)} = 1$ **[M1]**

$3x + 9 - 2x + 2 = (x-1)(x+3)$ **[M1]**

$x + 11 = x^2 + 2x - 3$ **[M1]**

$x^2 + x - 14 = 0$ established with no errors seen **[A1]**

(b) $\dfrac{-1 \pm \sqrt{1^2 - 4(1)(-14)}}{2(1)}$ **[M1]**

$\dfrac{-1 \pm \sqrt{57}}{2}$ **[M1]**

$3.27, -4.27$ **[A1]**

[Total Marks: 7]

32 (a) (i) Forms a correct initial equation involving time: $\dfrac{120}{x} - \dfrac{120}{x+10} = \dfrac{36}{60}$ **[M1]**

Simplifies the time difference to $\dfrac{3}{5}$ and multiplies by a common denominator: $600(x+10) - 600x = 3x(x+10)$ **[M1]**

$6000 = 3x^2 + 30x$ leading correctly to $x^2 + 10x - 2000 = 0$ with no errors **[A1]**

(ii) $(x+50)(x-40) = 0$ **[M1]**

$x = 40, x = -50$ **[A1]**

(b) Uses the quadratic formula with correct substitution: $x = \dfrac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(-5)}}{2(3)}$ **[M1]**

Simplifies to $\dfrac{7 \pm \sqrt{109}}{6}$ **[M1]**

$x = 2.91$ or $x = -0.57$ **[A1]**

[Total Marks: 8]

33 (a) Constructs initial time equation: $\dfrac{120}{x} + \dfrac{150}{x+10} = 4$ **[M1]**

Multiplies by common denominator $x(x+10)$ to reach $120(x+10) + 150x = 4x(x+10)$ **[M1]**

Expands brackets correctly: $120x + 1200 + 150x = 4x^2 + 40x$ **[M1]**

Rearranges to $4x^2 - 230x - 1200 = 0$ and divides by 2 to yield $2x^2 - 115x - 600 = 0$ with no errors seen **[A1]**

(b) Substitutes values into the quadratic formula $\dfrac{115 \pm \sqrt{(-115)^2 - 4(2)(-600)}}{4}$ **[M1]**

62.3 **[A1]**

(c) -1.84 **[B1]**

2.36 **[B1]**

[Total Marks: 8]

34 (a) $x(x-6) = 3(x+2)$ **[M1]**

$x^2 - 6x = 3x + 6$ **[M1]**

$x^2 - 9x - 6 = 0$ **[A1]**

Uses the quadratic formula, e.g., $x = \dfrac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(-6)}}{2(1)}$ **[M1]**

$\dfrac{9 \pm \sqrt{105}}{2}$ **[A1]**

(b) $y(3x+1) = 2x - 5$ **[M1]**

$3xy + y = 2x - 5$ **[M1]**

Isolates terms containing x on one side, e.g., $3xy - 2x = -y - 5$ or $2x - 3xy = y + 5$ **[M1]**

$\dfrac{-y-5}{3y-2}$ or $\dfrac{y+5}{2-3y}$ **[A1]**

[Total Marks: 9]

35 (a) $3(2x+1) + 2(x-1) = 2(x-1)(2x+1)$ **[M1]**

$6x + 3 + 2x - 2 = 4x^2 - 2x - 2$ **[M1]**

$4x^2 - 10x - 3 = 0$ **[A1]**

$\dfrac{10 \pm \sqrt{(-10)^2 - 4(4)(-3)}}{2(4)}$ **[M1]**

$2.77, -0.27$ **[A1]**

(b) $x^2 + (x+2)^2 = 340$ (or equivalent setup for consecutive even numbers) **[M1]**

$2x^2 + 4x - 336 = 0$ or $x^2 + 2x - 168 = 0$ **[A1]**

$(x+14)(x-12) = 0$ or correct application of quadratic formula **[M1]**

12 and 14 (both correct) **[A1]**

[Total Marks: 9]

36 (a) $\dfrac{100}{x} - \dfrac{100}{x+15} = \dfrac{1}{2}$ (or equivalent e.g., $\dfrac{100}{x} = \dfrac{100}{x+15} + 0.5$) **[M1]**

$200(x+15) - 200x = x(x+15)$ **[M1]**

$200x + 3000 - 200x = x^2 + 15x$ leading to $x^2 + 15x - 3000 = 0$ **[A1]**

(b) $\dfrac{-15 \pm \sqrt{15^2 - 4(1)(-3000)}}{2(1)}$ **[M2]**

Allow [M1] for correctly substituting values into the quadratic formula or evaluating the discriminant $\sqrt{12225}$

47.78 **[A1]**

-62.78 **[A1]**

(c) Speed of car $= 47.78 + 15 = 62.78$ km/h **[M1]**

Time $= \dfrac{100}{62.78} = 1.592...$ hours **[M1]**

1 hour 36 minutes **[A1]**

[Total Marks: 10]

37 (a) Multiplies by the denominator to obtain $y(x+4) = 3x - 2$ **[M1]**

Expands the bracket to obtain $xy + 4y = 3x - 2$ **[M1]**

Collects terms in x on one side, e.g., $xy - 3x = -4y - 2$ or $3x - xy = 4y + 2$ **[M1]**

Factorises x out and divides: $\dfrac{-4y-2}{y-3}$ or $\dfrac{4y+2}{3-y}$ **[A1]**

(b) Writes with common denominator or multiplies through to eliminate fractions:
$5(2x+1) - 3(x-2) = 2(x-2)(2x+1)$ **[M1]**

Correctly expands LHS: $10x + 5 - 3x + 6$ **[M1]**

Correctly expands RHS: $2(2x^2 - 3x - 2)$ or $4x^2 - 6x - 4$ **[M1]**

Rearranges to form a correct 3-term quadratic equation, e.g., $4x^2 - 13x - 15 = 0$ **[A1]**

Correctly substitutes into the quadratic formula: $\dfrac{-(-13) \pm \sqrt{(-13)^2 - 4(4)(-15)}}{2(4)}$ **[M1]**

$x = 4.15$ and $x = -0.90$ **[A1]**

[Total Marks: 10]

38 **(a)** $y(x+2) = 3x - 1$ **[M1]**

$yx + 2y = 3x - 1$ **[M1]**

$2y + 1 = 3x - yx$ or $yx - 3x = -2y - 1$ **[M1]**

$x = \dfrac{2y+1}{3-y}$ or $x = \dfrac{-2y-1}{y-3}$ **[A1]**

(b) Multiply throughout by $(x-1)(x+2)$: $2(x+2) + 3(x-1) = (x-1)(x+2)$ **[M1]**

Expand terms: $2x + 4 + 3x - 3 = x^2 + x - 2$ **[M1]**

Collect to form a quadratic equation: $x^2 - 4x - 3 = 0$ **[A1]**

Use the quadratic formula: $x = \dfrac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-3)}}{2(1)}$ **[M1]**

$x = \dfrac{4 \pm \sqrt{28}}{2}$ **[M1]**

$x = 4.65$ and $x = -0.65$ **[A1]**

[Total Marks: 10]

39 **(a)** $\dfrac{150}{x}$ **[B1]**

(b) $\dfrac{150}{x-15} - \dfrac{150}{x} = 0.5$ (or $\dfrac{30}{60}$ or $\dfrac{1}{2}$) **[M1]**

$150x - 150(x-15) = 0.5x(x-15)$ **[M1]**

$150x - 150x + 2250 = 0.5x^2 - 7.5x$ **[M1]**

$4500 = x^2 - 15x \implies x^2 - 15x - 4500 = 0$ with no errors seen **[A1]**

(c) $(x-75)(x+60) = 0$ or $\dfrac{-(-15) \pm \sqrt{(-15)^2 - 4(1)(-4500)}}{2(1)}$ **[M2]**

(or $M1$ for correct factors $(x+a)(x+b)$ where $ab = -4500$ or $a + b = -15$, or for correct substitution into quadratic formula)

75 (rejects negative solution -60) **[A1]**

(d) $P(q-1) = 2q + 3$ **[M1]**

$Pq - P = 2q + 3$ **[M1]**

$Pq - 2q = P + 3$ **[M1]**

$\dfrac{P+3}{P-2}$ or $\dfrac{-P-3}{2-P}$ **[A1]**

[Total Marks: 12]